RACE
Regular Analysis Through Continuous Exercise

Indefinite Integration : RACE-01

Subjective Type Question

Evaluate the following integrals :

1. **(i)** $\int 3x^{1/2}\left(1+x^{3/2}\right)dx$

MII130

(ii) $\int 2^x.e^x dx$

MII131

(iii) $\int \frac{e^{5\ell nx}-e^{4\ell nx}}{e^{3\ell nx}-e^{2\ell nx}}dx$

MII132

(iv) $\int (e^{a\ell nx}+e^{x\ell na})dx\ (a>0)$

MII133

2. **(i)** $\int \frac{\sin^3 x\cos^3 x+\cos^4 x\sin^2 x}{\sin^2 x\cos^3 x}dx$

MII134

(ii) $\int \frac{1+\cos^2 x}{1+\cos 2x}dx$

MII135

(iii) $\int \frac{\cos 2x}{\cos^2 x\sin^2 x}dx$

MII136

(iv) $I=\int \frac{dx}{\sin^2 x.\cos^2 x}$

MII137

3. **(i)** $\int \frac{\sin^3 x+\cos^3 x}{1-\sin x\cos x}dx$

MII138

(ii) $\int \frac{\sin^3 x+\cos^3 x}{\sin^2 x\cos^2 x}dx$

MII139

(iii) $\int\left(\sqrt{\frac{1+x}{1-x}}+\sqrt{\frac{1-x}{1+x}}\right)dx$

MII140

(iv) $\int \frac{2+2x^2-3\sqrt{1-x^2}}{\sqrt{\left(1-x^2\right)\left(x^4+2x^2+1\right)}}dx$

MII141

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4. **(i)** $\int \frac{2+3x^2}{x^2\left(1+x^2\right)}dx$

MII142

(ii) $\int \frac{1+\tan^2 x}{1+\cot^2 x}dx$

MII143

(iii) $\int \frac{x^6-1}{x^2+1}dx$

MII144

(iv) $\int \frac{\cos 2x-\cos 2\alpha}{\cos x-\cos \alpha}dx$

MII145

5. **(i)** $\int \frac{dx}{1-\cos x}$

MII146

(ii) $\int \frac{|x|\sqrt{x^2-1}+1+x^2}{|x|\sqrt{x^2-1}\left(1+x^2\right)}dx$

MII147

(iii) $\int \frac{\sin^6 x+\cos^6 x}{\sin^2 x.\cos^2 x}dx$

MII148

(iv) $\int \frac{\sin 2x+\sin 5x-\sin 3x}{\cos x+1-2\sin^2 2x}dx$

MII149

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6. **(i)** $\int \frac{dx}{(2x-3)^5}$

MII150

(ii) $\int \sqrt[5]{(8-3x)^6}dx$

MII151

(iii) $\int e^{-3x+1}dx$

MII152

(iv) $\int \frac{e^{2x}-1}{e^x}dx$

MII153

7. **(i)** $\int \frac{m}{\sqrt[3]{(a+bx)^2}}dx$

MII154

(ii) $\int \frac{2x^3+3x^2+4x+5}{2x+1}dx$

MII155

(iii) $\int \frac{2x+3}{3x+2}dx$

MII156

(iv) $\int (e^x+1)^3\,dx$

MII157

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8. **(i)** $\int \frac{x}{x^2+2x+1}dx$ **MII158**

(ii) $\int \cos x^\circ dx$ **MII159**

(iii) $\int \frac{1-\tan^2 x}{1+\tan^2 x}dx$ **MII160**

(iv) $\int \frac{1-\cos x}{1+\cos x}dx$ **MII161**

9. **(i)** $\int\left[\cos\left(2x-\frac{\pi}{4}\right)\right]^{-2}dx$ **MII162**

(ii) $\int \frac{\cos x-\sin x}{\cos x+\sin x}(2+2\sin 2x)dx$ **MII163**

(iii) $\int \cos x \sin 3x dx$ **MII164**

(iv) $\int (3\sin x\cos^2 x-\sin^3 x)dx$ **MII165**

10. **(i)** $\int \sin x\cos x\cos 2x\cos 4x\,dx$ **MII166**

(ii) $\int\left[\sin^2\left(\frac{9\pi}{8}+\frac{x}{4}\right)-\sin^2\left(\frac{7\pi}{8}+\frac{x}{4}\right)\right]dx$ **MII167**

(iii) $\int \frac{\cos 4x+1}{\cot x-\tan x}dx$ **MII168**

11. **(i)** $\int\left[\frac{\cot^2 2x-1}{2\cot 2x}-\cos 8x\cot 4x\right]dx$ **MII169**

(ii) $\int \cos x\cos 2x\cos 3x dx$ **MII170**

(iii) $\int 4\cos\frac{x}{2}.\cos x.\sin\frac{21}{2}x dx$ **MII171**

12. **(i)** $\int \frac{3}{(2x-1)(2x-3)}dx$ **MII172**

(ii) $\int \frac{x+1}{(4x-1)(3x-2)}dx$ **MII173**

(iii) $\int \frac{5x-2}{(2x+3)(3x-5)}dx$ **MII174**

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ANSWER KEY

1. **(i)** $2x^{3/2}+x^3+C$ **(ii)** $\frac{(2e)^x}{1+\ell n2}+C$ **(iii)** $\frac{x^3}{3}+C$ **(iv)** $\frac{x^{a+1}}{a+1}+\frac{a^x}{\ell na}+C$

2. **(i)** $-\cos x+\sin x+C$ **(ii)** $\frac{\tan x}{2}+\frac{x}{2}+C$ **(iii)** $-\cot x-\tan x+C$ **(iv)** $\tan x-\cot x+C$

3. **(i)** $-\cos x+\sin x+C$ **(ii)** $\sec x-\operatorname{cosec} x+C$ **(iii)** $2\sin^{-1}x+C$ **(iv)** $2\sin^{-1}x-3\tan^{-1}x+C$

4. **(i)** $\frac{-2}{x}+\tan^{-1}x+C$ **(ii)** $\tan x-x+C$ **(iii)** $\frac{x^5}{5}-\frac{x^3}{3}+x-2\tan^{-1}x+C$ **(iv)** $2\sin x+2\cos\alpha.x+C$

5. **(i)** $-\cot x-\operatorname{cosec} x+C$ **(ii)** $\tan^{-1}x+\sec^{-1}x+C$ **(iii)** $\tan x-\cot x-3x+C$ **(iv)** $-2\cos x+C$

6. **(i)** $\frac{-1}{8(2x-3)^4}+C$ **(ii)** $\frac{-5}{33}(8-3x)^{11/5}+C$ **(iii)** $\frac{e^{-3x+1}}{-3}+C$ **(iv)** $e^x+e^{-x}+C$

7. **(i)** $\frac{3m}{b}(a+bx)^{1/3}+C$ **(ii)** $\frac{x^3}{3}+\frac{x^2}{2}+\frac{3x}{2}+\frac{7\ell n(2x+1)}{4}+C$

(iii) $\frac{2x}{3}+\frac{5}{9}\ell n(3x+2)+C$ **(iv)** $\frac{e^{3x}}{3}+\frac{3e^{2x}}{2}+3e^x+x+C$

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8. **(i)** $\ell n(x+1)+\frac{1}{x+1}+C$ **(ii)** $\frac{180}{\pi}\sin x^\circ+C$ **(iii)** $\frac{\sin 2x}{2}+C$ **(iv)** $2\tan\frac{x}{2}-x+C$

9. **(i)** $\frac{\tan\left(2x-\frac{\pi}{4}\right)}{2}+C$ **(ii)** $\sin 2x+C$ **(iii)** $\frac{1}{2}\int\left(-\frac{\cos 4x}{4}-\frac{\cos 2x}{2}\right)+C$ **(iv)** $\frac{-\cos 3x}{3}+C$

10. **(i)** $\frac{1}{8}\left(-\frac{\cos 8x}{8}\right)+C$ **(ii)** $\frac{1}{\sqrt{2}}.2\left(-\cos\frac{x}{2}\right)+C$ **(iii)** $\frac{1}{2}\left(-\frac{\cos 4x}{4}\right)+C$

11. **(i)** $-\frac{\cos 8x}{8}+C$ **(ii)** $\frac{1}{4}\left(x+\frac{\sin 6x}{6}+\frac{\sin 4x}{4}+\frac{\sin 2x}{2}\right)+C$

(iii) $\frac{-\cos 12x}{12}-\frac{\cos 9x}{9}-\frac{\cos 11x}{11}-\frac{\cos 10x}{10}+C$

12. **(i)** $\frac{3}{4}\ell n\frac{2x-3}{2x-1}+C$ **(ii)** $\frac{\ell n|3x-2|}{3}-\frac{\ell n|4x-1|}{4}+C$ **(iii)** $\frac{\ell n|3x-5|}{3}+\frac{\ell n|2x+3|}{2}+C$

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RACE Regular Analysis Through Continuous Exercise

Indefinite Integration : RACE-02

Subjective Type Question

Evaluate the following integrals :

1. **(i)** $\int \frac{e^x dx}{e^x + 1}$ **MII175**

(ii) $\int \frac{d(1+x^2)}{\sqrt{1+x^2}}$ **MII176**

(iii) $\int \frac{d\left(\frac{x}{3}\right)}{\sqrt{1-\left(\frac{x}{3}\right)^2}}$ **MII177**

(iv) $\int \frac{d(1+\ell nx)}{\cos^2(1+\ell nx)}$ **MII178**

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2. **(i)** $\int \frac{d(\arcsin x)}{\arcsin x}$ **MII179**

(ii) $\int \frac{dx}{x\ell nx}$ **MII180**

(iii) $\int \frac{\sqrt{\ell nx}}{x} dx$ **MII181**

(iv) $\int \frac{e^x + 1}{e^x + x} dx$ **MII182**

3. **(i)** $\int \frac{e^x + \cos x}{e^x + \sin x} dx$ **MII183**

(ii) $\int 2x\sqrt{x^2+1}\, dx$ **MII184**

(iii) $\int \frac{(2x-3)dx}{x^2 - 3x + 8}$ **MII185**

(iv) $\int e^{-x^3} x^2 dx$ **MII186**

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4. **(i)** $\int \frac{(6x-5)dx}{2\sqrt{3x^2-5x+6}}$

MII187

(ii) $\int \frac{e^{\sqrt{x}}}{\sqrt{x}}dx$

MII188

(iii) $\int \frac{x^2dx}{x^3+1}$

MII189

(iv) $\int \frac{x-(x+1)\ell n(x+1)}{x(x+1)\ell n(x+1)}dx$

MII190

5. **(i)** $\int \frac{e^{2x}dx}{e^{2x}+a^2}$

MII191

(ii) $\int (x+1)\sqrt{x^2+2x}\,dx$

MII192

(iii) $\int \frac{\sqrt{x}dx}{1+x^{\frac{3}{2}}}$

MII193

(iv) $\int \frac{2x^5-3x^2}{1+3x^3-x^6}dx$

MII194

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6. **(i)** $\int e^{2x^2+\ell nx}\,dx$

MII195

(ii) $\int x^2\sqrt[5]{x^3+2}\,dx$

MII196

(iii) $\int \frac{x^4dx}{\sqrt{4+x^5}}$

MII197

(iv) $\int \frac{(\arctan x)^n}{1+x^2}dx$

MII198

7. **(i)** $\int \frac{dx}{x\ln x\ln(\ln x)}$

MII199

(ii) $\int \frac{x}{\sqrt{x+2}}dx$

MII200

(iii) $\int \frac{x}{2x+1}dx$

MII201

(iv) $\int \frac{e^x-1}{e^x+1}dx$

MII202

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8. **(i)** $\int \frac{x^4}{1-x}dx$

MII203

(ii) $\int \frac{x^3dx}{\sqrt{1+x^2}}$

MII204

(iii) $\int x^x \ell n(ex)dx$

MII205

(iv) $\int \frac{dx}{1+\sqrt{x+1}}$

MII206

9. **(i)** $\int \frac{\sin 2x}{1+\cos^2 x}dx$

MII207

(ii) $\int \frac{x^5+3x^4-x^3+8x^2-x+8}{x^2+1}dx$

MII208

(iii) $\int \frac{(\sqrt{x}+1)(x^2-\sqrt{x})}{x\sqrt{x}+x+\sqrt{x}}dx$

MII209

(iv) $\int \frac{dx}{1+\sqrt[3]{x+1}}$

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MII210

10. **(i)** $\int\left(\frac{x-1}{x^2+1}\right)^2 dx$

MII211

(ii) $\int \frac{dx}{(x+\sqrt{x^2-1})^2}$

MII212

(iii) $\int \frac{(\sqrt{x}+1)dx}{\sqrt{x}\left(\sqrt[3]{x}+1\right)}$

MII213

(iv) $\int \frac{x^2dx}{(a+bx)^2}$

MII214

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ANSWER KEY

1. **(i)** $\ell n(e^x+1)+C$ **(ii)** $2\sqrt{1+x^2}+C$ **(iii)** $\arcsin\frac{x}{3}+C$ **(iv)** $\tan(1+\ell nx)+C$

2. **(i)** $\ell n|\arcsin x|+C$ **(ii)** $\ell n|\ell nx|+C$ **(iii)** $\frac{2}{3}\sqrt{(\ell nx)^3}+C$ **(iv)** $\ell n|e^x+x|+C$

3. **(i)** $\ell n\left(e^x+\sin x\right)+c$ **(ii)** $\frac{2}{3}\sqrt{(x^2+1)^3}+C$ **(iii)** $\ell n(x^2-3x+8)+C$ **(iv)** $C-\frac{1}{3}e^{-x^3}$

4. **(i)** $\sqrt{3x^2-5x+6}+c$ **(ii)** $2e^{\sqrt{x}}+c$ **(iii)** $\frac{1}{3}\ell n\left(x^3+1\right)+c$ **(iv)** $\ell n\left(\frac{\ell n(1+x)}{x}\right)+c$

5. **(i)** $\frac{1}{2}\ell n(e^{2x}+a^2)+C$ **(ii)** $\frac{1}{3}\sqrt{(x^2+2x)^3}+c$ **(iii)** $\frac{2}{3}\ell n(1+x^{\frac{3}{2}})+c$ **(iv)** $-\frac{1}{3}\left(1+3x^3-x^6\right)+C$

6. **(i)** $\frac{1}{4}e^{2x^2}+c$ **(ii)** $\frac{5}{18}\sqrt[5]{(x^3+2)^6}+C$ **(iii)** $\frac{2}{5}\sqrt{.4+x^5}+C$ **(iv)** $\frac{(\arctan x)^3}{3}+C$

7. **(i)** $\int\frac{dt}{t}=\ell nt=\ell n\left(\ell n\left(\ell nx\right)\right)+c$ **(ii)** $\frac{2}{3}(x+2)^{3/2}-4(x+2)^{1/2}+C$

(iii) $\frac{1}{2}\left[x-\frac{\ell n(2x+1)}{2}\right]+C$ **(iv)** $2\ell n(e^{x/2}+e^{-x/2})+C$

8. **(i)** $C-\frac{1}{4}x^4-\frac{1}{3}x^3-\frac{1}{2}x^2-x-\ell n|1-x|$ **(ii)** $\frac{1}{3}\left(1+x^2\right)^{\frac{3}{2}}-\left(1+x^2\right)^{\frac{1}{2}}+C$

(iii) x^x+C **(iv)** $2\left[\sqrt{x+1}-\ell n(1+\sqrt{x+1})\right]+C$

9. **(i)** $-\ell n\left(1+\cos^2 x\right)+c$ **(ii)** $\frac{x^4}{4}+x^3-x^2+5x+\frac{1}{2}\ell n(x^2+1)+3\tan^{-1}x+C$

(iii) $\frac{x^2}{2}-x+C$ **(iv)** $\frac{3}{2}(x+1)^{3/2}-3(x+1)^{1/3}+3\ell n\left|1+\sqrt[3]{x+1}\right|+C$

10. **(i)** $\tan^{-1}x-\frac{1}{x^2+1}+c$ **(ii)** $\frac{2}{3}[x^3-\sqrt{(x^2-1)^3}]-x+C$

(iii) $6\left[\frac{t^4}{4}-\frac{t^2}{2}+t+\frac{1}{2}l\,n(1+t^2)-\tan^{-1}t\right]+C$ where $t=x^{1/6}$

(iv) $\frac{1}{b^2}x-\frac{a}{b^3}\ell n\left(bx+a^2\right)+\frac{a^2}{b^2}\left(-\frac{1}{bx+a}\right)\frac{1}{b}+c$

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Regular Analysis Through Continuous Exercise

Indefinite Integration : RACE-03

Subjective Type Question

Evaluate the following integrals :

1. **(i)** $\int \frac{\sin x dx}{(1-\cos x)^2}$ **MII215**

(ii) $\int \frac{\cos x\, dx}{\sqrt[3]{\sin^2 x}}$ **MII216**

(iii) $\int \frac{dx}{\cos^2 x\sqrt{1+\tan x}}$ **MII217**

(iv) $\int \left(\tan^3 x - x\tan^2 x\right)dx$ **MII218**

2. **(i)** $\int \frac{e^{\cot x}+e^{-\cot x}}{1-\cos^2 x}dx$ **MII219**

(ii) $\int \frac{1-\sin x}{\cos x}dx$ **MII220**

(iii) $\int \tan^3 x\, dx$ **MII221**

(iv) $\int \frac{\sin 2x}{\left(a\sin^2 x+b\cos^2 x\right)^2}dx$ **MII222**

3. **(i)** $\int \frac{\sin^3 x}{\cos x}dx$ **MII223**

(ii) $\int \frac{\cos^3 x dx}{\sin^4 x}$ **MII224**

(iii) $\int \frac{\sin^3 \alpha}{\sqrt{\cos\alpha}}d\alpha$ **MII225**

(iv) $\int \frac{1+\sin x}{1-\sin x}dx$ **MII226**

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4. **(i)** $\int\left(\frac{x\cos x-\sin x}{x\sin x}\right)dx$

MII227

(ii) $\int\cot\left(2x+1\right)dx$

MII228

(iii) $\int\tan 3x\,dx$

MII229

(iv) $\int\frac{\sin^2 x+\cos x}{\sin x\cos x}dx$

MII230

5. **(i)** $\int\frac{\sin x+\cos x}{\sin x\cos x}dx$

MII231

(ii) $\int\frac{\sin 5x}{\cos 2x\cos 3x}dx$

MII232

(iii) $\int\frac{d\phi}{\sqrt{3}\cos\phi+\sin\phi}$

MII233

(iv) $\int\frac{dx}{\cot\frac{x}{2}.\cot\frac{x}{3}.\cot\frac{x}{6}}$

MII234

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6. **(i)** $\int\sin^2 x\cos^5 x\,dx$

MII235

(ii) $\int\cos^3 x dx$

MII236

(iii) $\int\sin^4 x dx$

MII237

(iv) $\int\frac{dx}{\sqrt[4]{\sin^3 x\cos^5 x}}$

MII238

7. **(i)** $\int\frac{\sin^2 x}{\cos^4 x}dx$

MII239

(ii) $\int\frac{\sqrt{\sin x}dx}{\cos^{5/2} x}$

MII240

(iii) $\int\frac{dx}{\cos^4 x}$

MII241

(iv) $\int\frac{dx}{\cos x.\sin^3 x}$

MII242

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8. **(i)** $\int (\sin x)^{-11/3} (\cos x)^{-1/3}\, dx$

MII243

(ii) $\int \sqrt[3]{\frac{\sin^2 x}{\cos^{14} x}}\, dx$

MII244

(iii) $\int \frac{dx}{\cos^3 x \sqrt{\sin 2x}}$

MII245

9. **(i)** $\int \frac{dx}{1+9x^2}$

MII246

(ii) $\int \frac{dx}{4x^2-9}$

MII247

(iii) $\int \frac{dx}{2-3x^2}$

MII248

10. **(i)** $\int \frac{e^x dx}{e^{2x}+4}$

MII249

(ii) $\int \frac{x^2}{9+16x^6} dx$

MII250

(iii) $\int \frac{dx}{(x-1)^2+4}$

MII251

(iv) $\int \frac{dx}{x^2+2x+3}$

MII252

11. **(i)** $\int \frac{dx}{4x^2+12x-7}$

MII253

(ii) $\int \frac{dx}{x^2+x+1}$

MII254

(iii) $\int \frac{\cos x\, dx}{3\sin^2 x+4\sin x+5}$

MII255

(iv) $\int \frac{x\, dx}{x^4-3x^2+2}$

MII256

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ANSWER KEY

1. **(i)** $\frac{1}{\cos x-1}+c$ **(ii)** $3\sqrt[3]{\sin x}+C$ **(iii)** $2\sqrt{1+\tan x}+C$ **(iv)** $=\frac{(\tan x-x)^2}{2}+c$

2. **(i)** $-e^{\cot x}+e^{-\cot x}+C$ **(ii)** $\ell n(1+\sin x)+C$

(iii) $\frac{1}{2}\tan^2 x+\ell n|\cos x|+C$ **(iv)** $\frac{-1}{a+b}\frac{1}{\left(a\sin^2 x+b\cos^2 x\right)}+c$

3. **(i)** $\frac{\cos^2 x}{2}-\ell n|\cos x|+C$ **(ii)** $\frac{1}{\sin x}-\frac{1}{3\sin^3 x}+C$

(iii) $2\sqrt{\cos\alpha}\left(\frac{\cos^2\alpha}{5}-1\right)+C$ **(iv)** $2\tan\left(\frac{x}{2}+\frac{\pi}{4}\right)-x+C$

4. **(i)** $\ell n\left(\frac{\sin x}{x}\right)+C$ **(ii)** $\frac{1}{2}\ell n|\sin(2x+1)|+C$ **(iii)** $C-\frac{1}{3}\ell n|\cos 3x|$ **(iv)** $\ell n\left|\frac{\sec x}{\cos ecx+\cot x}\right|+c$

5. **(i)** $\ell n\left|\frac{\sec x+\tan x}{\cos ecx+\cot x}\right|+c$ **(ii)** $\frac{1}{3}\ell n\sec 3x+\frac{1}{2}\ell n\sec 2x+c$

(iii) $\frac{1}{2}\ell n\left|\tan\left(\frac{\phi}{2}+\frac{\pi}{6}\right)\right|+c$ **(iv)** $2\ell n\left(\sec\frac{x}{2}\right)-3\ell n\left(\sec\frac{x}{3}\right)-6\ell n\left(\sec\frac{x}{6}\right)+C$

6. **(i)** $\frac{1}{3}\sin^3 x-\frac{2}{5}\sin^5 x+\frac{1}{7}\sin^7 x+C$ **(ii)** $\sin x-\frac{1}{3}\sin^3 x+C$

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(iii) $\frac{3}{8}x-\frac{1}{4}\sin 2x+\frac{1}{32}\sin 4x+C$ **(iv)** $4\sqrt[4]{\tan x}+c$

7. **(i)** $\frac{1}{3}\tan^3 x+C$ **(ii)** $\frac{2}{3}\tan^{3/2} x+C$ **(iii)** $\tan x+\frac{\tan^3 x}{3}+c$ **(iv)** $\ell n|\tan x|-\frac{1}{2\sin^2 x}+c$

8. **(i)** $\frac{-3}{8}\left(\frac{1+4\tan^2 x}{\tan^{\frac{2}{3}} x}\right)+c$ **(ii)** $\frac{3}{55}\tan^{\frac{5}{3}} x\left(11+5\tan^2 x\right)+c$ **(iii)** $\frac{\sqrt{2}}{5}\sqrt{\tan x}\left(5+\tan^2 x\right)+c$

9. **(i)** $\frac{1}{3}\tan^{-1}3x+c$ **(ii)** $\frac{1}{12}\ell n\left|\frac{2x-3}{2x+3}\right|+C$ **(iii)** $\frac{1}{2\sqrt{6}}\ell n\left|\frac{\sqrt{2}+x\sqrt{3}}{\sqrt{2}-x\sqrt{3}}\right|+C$

10. **(i)** $\frac{1}{2}\text{arc}\ \tan\frac{e^x}{2}+C$ **(ii)** $\frac{1}{36}\tan^{-1}\left(\frac{4x^3}{3}\right)+C$ **(iii)** $\frac{1}{2}\arctan\frac{x-1}{2}+C$ **(iv)** $\frac{1}{\sqrt{2}}\arctan\frac{x+1}{\sqrt{2}}+C$

11. **(i)** $\frac{1}{16}\ell n\left|\frac{2x-1}{2x+7}\right|+c$ **(ii)** $\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+C$

(iii) $\frac{1}{\sqrt{11}}\tan^{-1}\frac{3t+2}{\sqrt{11}}+c$ **(iv)** $\ell n\sqrt{\frac{x^2-2}{x^2-1}}+C$

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RACE Regular Analysis Through Continuous Exercise

Indefinite Integration : RACE-04

Subjective Type Question

Evaluate the following integrals :

1. **(i)** $\int \frac{dx}{\sqrt{1-25x^2}}$ **MII257**

(ii) $\int \frac{dx}{\sqrt{4-9x^2}}$ **MII258**

2. **(i)** $\int \frac{dx}{\sqrt{4x^2+9}}$ **MII259**

(ii) $\int \frac{x^{\frac{5}{2}}}{\sqrt{1+x^7}}dx$ **MII260**

3. **(i)** $\int \frac{dx}{\sqrt{x^2+2x+10}}$ **MII261**

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(ii) $\int \frac{dx}{\sqrt{9x^2+12x-5}}$ **MII262**

4. **(i)** $\int \frac{dx}{\sqrt{1-(2x+3)^2}}$ **MII263**

(ii) $\int \frac{dx}{\sqrt{x-x^2}}$ **MII264**

5. **(i)** $\int \frac{dx}{\sqrt{8+6x-9x^2}}$ **MII265**

(ii) $\int \frac{dx}{\sqrt{1+e^{-2x}}}$ **MII266**

6. **(i)** $\int \frac{dx}{\sqrt{9+16\sec^2 x}}$ **MII267**

(ii) $\int \frac{2x+3}{\sqrt{1+x^2}}dx$ **MII268**

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7. **(i)** $\int\sqrt{\frac{1+x}{1-x}}dx$ **MII269**

(ii) $\int\sqrt{\frac{x-3}{2-x}}dx$ **MII270**

8. **(i)** $\int\sqrt{6-x^2}dx$ **MII271**

(ii) $\int\sqrt{9x^2-5}\,dx$ **MII272**

9. **(i)** $\int\sqrt{1-4x-x^2}dx$ **MII273**

(ii) $\int\sqrt{1+3x-x^2}\,dx$ **MII274**

10. **(i)** $\int\sqrt{x^2+4x+6}\,dx$ **MII275**

(ii) $\int\sqrt{1-\sin^4 x}dx$ **MII276**

ANSWER KEY

1. **(i)** $\frac{1}{5}\text{arc}\,\sin 5x+C$ **(ii)** $\frac{1}{3}\text{arc}\,\sin\frac{3x}{2}+C$

2. **(i)** $\frac{1}{2}\ell n\left(2x+\sqrt{4x^2+9}\right)+c$ **(ii)** $\frac{2}{7}\ell n\left|x^{\frac{7}{2}}+\sqrt{1+x^7}\right|+c$

3. **(i)** $\ell n\left|x+1+\sqrt{x^2+2x+10}\right|+c$ **(ii)** $\frac{1}{3}\ell n\left|3x+2+\sqrt{9x^2+12x-5}\right|+c$

4. **(i)** $\frac{1}{2}\arcsin(2x+3)+C$ **(ii)** $2\arcsin\sqrt{x}+C$

5. **(i)** $\frac{1}{3}\arcsin\frac{3x-1}{3}+C$ **(ii)** $\ell n\left|e^x+\sqrt{e^{2x}+1}\right|+c$

6. **(i)** $\frac{1}{3}\sin^{-1}\frac{3\sin x}{5}+c$ **(ii)** $2\sqrt{1+x^2}+3\ell n(x+\sqrt{1+x^2})+c$

7. **(i)** $\text{arc}\,\sin x-\sqrt{1-x^2}+C$ **(ii)** $\sqrt{(x-2)(3-x)}-\sin^{-1}\sqrt{3-x}+C$

8. **(i)** $\frac{x}{2}\sqrt{6-x^2}+\frac{6}{2}\sin^{-1}\frac{x}{\sqrt{6}}+c$ **(ii)** $=\frac{x}{2}\sqrt{9x^2-5}-\frac{5}{6}\ell n\left|x+\sqrt{x^2-\frac{5}{9}}\right|+c$

9. **(i)** $\frac{5}{2}\sin^{-1}\left(\frac{x+2}{\sqrt{5}}\right)+\frac{x+2}{2}\sqrt{1-4x-x^2}+C$ **(ii)** $\frac{2x-3}{4}\sqrt{1+3x-x^2}+\frac{13}{8}\sin^{-1}\frac{2x-3}{\sqrt{13}}+c$

10. **(i)** $\frac{(x+2)}{2}\sqrt{x^2+4x+6}+\log\left|x+2+\sqrt{x^2+4x+6}\right|+C$

(ii) $=\frac{\sin x}{2}\sqrt{1+\sin^2 x}+\frac{1}{2}\ell n\left|\sin x+\sqrt{1+\sin^2 x}\right|+C$

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RACE Regular Analysis Through Continuous Exercise

Indefinite Integration : RACE-05

Subjective Type Question

Evaluate the following integrals :

1. **(i)** $\int \frac{(x+2)dx}{x^2+2x+2}$

MII277

(ii) $\int \frac{(3x-1)dx}{\sqrt{x^2+2x+2}}$

MII278

2. **(i)** $\int \frac{5x+4}{\sqrt{x^2+4x+1}}dx$

MII279

(ii) $\int x\sqrt{x+x^2}dx$

MII280

3. **(i)** $\int (x+1)\sqrt{2x^2+3}\,dx$

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MII281

(ii) $\int \frac{x^2+2x+2}{x^2+1}dx$

MII282

4. **(i)** $\int \frac{2x^2+5x+6}{\sqrt{x^2+1}}dx$

MII283

(ii) $\int \frac{x^2+1}{x^4-x^2+1}dx$

MII284

5. **(i)** $\int \frac{1}{1+x^4}dx$

MII285

(ii) $\int \frac{(e^{3x}+e^x)dx}{e^{4x}-e^{2x}+1}$

MII286

6. **(i)** $\int \frac{(2x^2-5)dx}{x^4-5x^2+6}$

MII287

(ii) $\int \left(\sqrt{\tan x}-\sqrt{\cot x}\right)dx$

MII288

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7. **(i)** $\int \frac{x-1}{(x+1)\sqrt{x^3+x^2+x}}dx$

MII289

(ii) $\int \frac{2+3x^2}{1+x^4}dx$

MII290

8. **(i)** $\int \sqrt{\tan x}dx$

MII291

(ii) $\int \frac{x}{(x-3)\sqrt{x+1}}dx$

MII292

9. **(i)** $\int \frac{1}{(x^2-4)\sqrt{2x+1}}dx$

MII293

(ii) $\int \frac{dx}{(1-x^2)\sqrt{2+x}}$

MII294

10. **(i)** $\int \frac{dx}{x\sqrt{x^2+4}}$

MII295

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(ii) $\int \frac{\left(x+\sqrt{1+x^2}\right)^{15}}{\sqrt{1+x^2}}.dx$

MII296

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ANSWER KEY

1. **(i)** $\frac{1}{2}\ell n(x^2+2x+2)+\arctan(x+1)+c$ **(ii)** $3\sqrt{x^2+2x+2}-4\ell n(x+1+\sqrt{x^2+2x+2})+c$

2. **(i)** $5\sqrt{x^2+4x+1}-6\ell n\left|x+2+\sqrt{x^2+4x+1}\right|+C$

(ii) $\frac{\left(x+x^2\right)^{\frac{3}{2}}}{3}-\frac{1}{2}\left(\frac{x+\frac{1}{2}}{2}\sqrt{x^2+x}-\frac{1}{4.2}\ell n\left(x+\frac{1}{2}+\sqrt{x^2+x}\right)\right)+c$

3. **(i)** $\frac{1}{6}\left(2x^2+3\right)^{\frac{3}{2}}+\frac{x}{2}\sqrt{2x^2+3}+\frac{3\sqrt{2}}{4}\ell n\left|x+\sqrt{x^2+\frac{3}{2}}\right|+c$

(ii) $x+\ell n\left|x^2+1\right|+\tan^{-1}x+c$

4. **(i)** $x\sqrt{x^2+1}+5\ell n\left|x+\sqrt{x^2+1}\right|+5\sqrt{x^2+1}+c$ **(ii)** $\tan^{-1}\left(\frac{x^2-1}{x}\right)+C$

5. **(i)** $\frac{1}{2\sqrt{2}}\left[\tan^{-1}\left(\frac{x^2-1}{\sqrt{2}x}\right)-\frac{1}{2}\ell n\frac{x^2-\sqrt{2}x+1}{x^2+\sqrt{2}x+1}\right]$ **(ii)** $\tan^{-1}\left(e^x-e^{-x}\right)+c$

6. **(i)** $\frac{1}{2\sqrt{2}}\ell n\left|\frac{x-\sqrt{2}}{x+\sqrt{2}}\right|+\frac{1}{2\sqrt{3}}\ell n\left|\frac{x-\sqrt{3}}{x+\sqrt{3}}\right|+c$ **(ii)** $\frac{1}{\sqrt{2}}\ell n\left|\frac{\sqrt{\tan x}+\sqrt{\cot x}-\sqrt{2}}{\sqrt{\tan x}+\sqrt{\cot x}+\sqrt{2}}\right|+c$

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7. **(i)** $2\tan^{-1}\sqrt{x+\frac{1}{x}+1}+C$ **(ii)** $\frac{5}{2\sqrt{2}}\tan^{-1}\frac{x-\frac{1}{x}}{\sqrt{2}}+\frac{1}{4\sqrt{2}}\ell n\left|\frac{x+\frac{1}{x}-\sqrt{2}}{x+\frac{1}{x}+\sqrt{2}}\right|+C$

8. **(i)** $\frac{1}{\sqrt{2}}\tan^{-1}\frac{\sqrt{\tan x}-\sqrt{\cot x}}{\sqrt{2}}+\frac{1}{2\sqrt{2}}\ell n\left|\frac{\sqrt{\tan x}+\sqrt{\cot x}-\sqrt{2}}{\sqrt{\tan x}+\sqrt{\cot x}+\sqrt{2}}\right|+C$

(ii) $2\sqrt{x+1}+\frac{3}{2}\ell n\left|\frac{\sqrt{x+1}-2}{\sqrt{x+1}+2}\right|+C$

9. **(i)** $\frac{1}{4\sqrt{5}}\ell n\left|\frac{\sqrt{2x+1}-\sqrt{5}}{\sqrt{2x+1}+\sqrt{5}}\right|-\frac{1}{2\sqrt{3}}\tan^{-1}\frac{\sqrt{2x+1}}{\sqrt{3}}+C$

(ii) $\frac{1}{2\sqrt{3}}\ell n\left(\frac{\sqrt{x+2}+\sqrt{3}}{\sqrt{x+2}-\sqrt{3}}\right)+\frac{1}{2}\ell n\left(\frac{\left|\sqrt{x+2}-1\right|}{\sqrt{x+2}+1}\right)+c$

10. **(i)** $\frac{-1}{2}\ell n\left|\frac{1}{x}+\sqrt{\frac{1}{x^2}+\frac{1}{4}}\right|+c$ **(ii)** $\frac{\left(x+\sqrt{1+x^2}\right)^{15}}{15}+c$

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RACE
Regular Analysis Through Continuous Exercise

Indefinite Integration : RACE-06

Subjective Type Question

Evaluate the following integrals :

1. **(i)** $\int \frac{dx}{x^n\left(1+x^n\right)^{1/n}}$

MII297

(ii) $\int \frac{dx}{x(x^2+1)}$

MII298

2. **(i)** $\int \frac{dx}{x^2(x^3+1)^{2/3}}$

MII299

(ii) $\int \frac{dx}{x^3(x^3+1)^{1/3}}$

MII300

3. **(i)** $\int \frac{1+x^4}{(1-x^4)^{3/2}}dx$

MII301

(ii) $\int \frac{2x-4}{x^2\sqrt{x^2-2x+2}}$

MII302

4. **(i)** $\int \frac{dx}{x^2(x^5+1)^{4/5}}$

MII303

(ii) $\int \frac{x\sin\alpha+2}{(x^2+x\sin\alpha+1)^{3/2}}dx$

MII304

5. **(i)** $\int \frac{\left(x^4-x\right)^{1/4}}{x^5}dx$

MII305

(ii) $\int \frac{x^2-1}{x^3\sqrt{2x^4-2x^2+1}}dx$

MII306

6. **(i)** $\int \frac{dx}{x\left(x^4+1\right)\left(x^4+2\right)}$

MII307

(ii) $\int x^{1/3}\left[1+x^{1/3}+x^{2/3}\right]\left[2x^{2/3}+3x^{1/3}+6\right]^3 dx$

MII308

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7. **(i)** $\int \frac{\left(2x^{12}+5x^{9}\right)}{\left(x^{5}+x^{3}+1\right)^{3}}dx$

MII309

(ii) $\int\left(x^{7m}+x^{2m}+x^{m}\right)\left(2x^{6m}+7x^{m}+14\right)^{1/m}dx$

MII310

8. **(i)** $\int \frac{5x^{8}+7x^{6}}{\left(x^{2}+1+2x^{7}\right)^{2}}dx$

MII311

(ii) $\int \frac{\cos 2x\,dx}{1+\sin x\,\cos x}$

MII312

9. **(i)** $\int \frac{\sin 2x\,dx}{4-\cos^{2}2x}$

MII313

(ii) $\int \frac{dx}{1+4\sin^{2}x}$

MII314

10. **(i)** $\int \frac{\sin x}{\sin x+\cos x}dx$

MII315

(ii) $\int \frac{\sin x}{\sin 3x}dx$

MII316

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11. **(i)** $\int \frac{dx}{4-3\cos^{2}x+5\sin^{2}x}$

MII317

(ii) $\int \frac{dx}{3\sin^{2}x+\sin x\cos x+1}$

MII318

12. **(i)** $\int \frac{dx}{3+\sin x}$

MII319

(ii) $\int \frac{dx}{5-4\sin x+3\cos x}$

MII320

13. **(i)** $\int\left(\sqrt{\tan x}+\sqrt{\cot x}\right)dx$

MII321

(ii) $\int \frac{3\sin x+2\cos x}{3\cos x+2\sin x}dx$

MII322

14. **(i)** $\int \frac{\sec^{2}x}{\left(\sec x+\tan x\right)^{9/2}}dx$

MII323

(ii) $\int \frac{dx}{\sin^{4}x+\cos^{4}x}$

MII324

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ANSWER KEY

1. **(i)** $\frac{1}{(1-n)}\left(\frac{1}{x^n}+1\right)^{\frac{-1}{n}+1}+c$ **(ii)** $-\frac{1}{2}\ell n\left(1+\frac{1}{x^2}\right)+c$

2. **(i)** $-\left(1+\frac{1}{x^3}\right)^{\frac{1}{3}}+c$ **(ii)** $-\frac{1}{2}\left(1+\frac{1}{x^3}\right)^{\frac{2}{3}}+c$

3. **(i)** $\frac{1}{\left(\frac{1}{x^2}-x^2\right)^{1/2}}+C$ **(ii)** $\frac{2\sqrt{x^2-2x+2}}{x}+c$

4. **(i)** $-\left(1+\frac{1}{x^5}\right)^{\frac{1}{5}}+c$ **(ii)** $\frac{2}{\left(1+\frac{\sin\alpha}{x}+\frac{1}{x^2}\right)^{\frac{1}{2}}}+c$

5. **(i)** $\frac{4}{15}\left(1-\frac{1}{x^3}\right)^{\frac{5}{4}}+c$ **(ii)** $\frac{1}{2}\frac{\sqrt{2x^4-2x^2+1}}{x^2}+c$

6. **(i)** $-\frac{1}{8}\left(\ell n\frac{\left(x^4+1\right)^2}{x^8+2x^4}\right)+c$ **(ii)** $\frac{\left(2x+3x^{\frac{2}{3}}+6x^{\frac{1}{3}}\right)^4}{8}+c$

7. **(i)** $\frac{x^{10}}{2\left(x^5+x^3+1\right)^2}+c$ **(ii)** $\frac{1}{14(m+1)}\left(2x^{7m}+7x^{2m}+14x^m\right)^{\frac{m+1}{m}}+c$

8. **(i)** $\frac{1}{\left(\frac{1}{x^5}+\frac{1}{x^7}+2\right)}+c$ **(ii)** $\ell n(2+\sin 2x)+c$

9. **(i)** $-\frac{1}{8}\ell n\left|\frac{2+\cos 2x}{2-\cos 2x}\right|+c$ **(ii)** $\frac{1}{\sqrt{5}}\tan^{-1}\left(\sqrt{5}\tan x\right)+c$

10. **(i)** $\frac{1}{2}x-\frac{1}{2}\ell n|\sin x+\cos x|+c$ **(ii)** $-\frac{1}{2\sqrt{3}}\ell n\left|\frac{\tan x-\sqrt{3}}{\tan x+\sqrt{3}}\right|+c$

11. **(i)** $\frac{1}{3}\tan^{-1}(3\tan x)+c$ **(ii)** $\frac{2}{\sqrt{15}}\tan^{-1}\left(\frac{8\tan x+1}{\sqrt{15}}\right)+c$

12. **(i)** $\frac{1}{\sqrt{2}}\tan^{-1}\frac{3\tan\frac{x}{2}+1}{2\sqrt{2}}+c$ **(ii)** $\frac{-1}{\tan\frac{x}{2}-2}+C$

13. **(i)** $\sqrt{2}\sin^{-1}(\sin x-\cos x)+c$ **(ii)** $\frac{7}{13}\times\frac{-5}{13}\ell n|3\cos x+2\sin x|+c$

14. **(i)** $-\frac{1}{(\sec x+\tan x)^{\frac{11}{2}}\left(\frac{1}{11}+\frac{(\sec x+\tan x)^2}{7}\right)}+c$ **(ii)** $\frac{1}{\sqrt{2}}\tan^{-1}\frac{\left(\tan^{-1}x-\cot x\right)}{\sqrt{2}}+C$

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RACE Regular Analysis Through Continuous Exercise

Indefinite Integration : RACE-07

Subjective Type Question

Evaluate the following integrals :

1. **(i)** $\int \frac{3x+2}{(x+1)(x+3)}dx$

MII325

(ii) $\int \frac{1}{(x+1)(x+2)(x+3)}dx$

MII326

2. **(i)** $\int \frac{dx}{6x^3-7x^2-3x}$

MII327

(ii) $\int \frac{xdx}{(x+1)^3}$

MII328

3. **(i)** $\int \frac{4x+3}{(x-2)^3}dx$

MII329

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(ii) $\int \frac{x^2dx}{(x+2)^2(x+4)^2}$

MII330

4. **(i)** $\int \frac{x^2-1}{(x+1)(x+2)^2}dx$

MII331

(ii) $\int \frac{1}{(x^2+1)(x+3)}dx$

MII332

5. **(i)** $\int \frac{x^2dx}{1-x^4}$

MII333

(ii) $\int \frac{3x+2}{(x+1)^2(x+2)}dx$

MII334

6. **(i)** $\int \frac{xdx}{x^3-1}$

MII335

(ii) $\int \frac{x^4dx}{x^{15}-1}$

MII336

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7. **(i)** $\int x\ell \text{nx}\, dx$

MII337

(ii) $\int xe^{-x}dx$

MII338

8. **(i)** $\int 2x^3 e^{x^2} dx$

MII339

(ii) $\int x^3 \sin(x^2)dx$

MII340

9. **(i)** $\int \frac{\ell n^3 x}{x^2} dx$

MII341

(ii) $\int \frac{\arcsin\sqrt{x}}{\sqrt{1-x}} dx$

MII342

10. **(i)** $\int \tan^{-1} x dx$

MII343

(ii) $\int \frac{x^2 \tan^{-1} x}{1+x^2} dx$

MII344

11. **(i)** $\int \arctan\sqrt{x}dx$

MII345

(ii) $\int \ell n(x+\sqrt{1+x^2})dx$

MII346

12. **(i)** $\int \sec^3 x dx$

MII347

(ii) $\int \frac{\sin^{-1} x}{\left(1-x^2\right)^{3/2}} dx$

MII348

13. **(i)** $\int x^3 \sin x dx$

MII349

(ii) $\int \left(x^2-2x+3\right)\cos 3x dx$

MII350

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ANSWER KEY

1. **(i)** $\frac{-1}{2}\ell n|x+1|+\frac{7}{2}\ell n|x+3|+C$ **(ii)** $\frac{1}{2}\ell n|x+1|-\ell n|x+2|+\frac{1}{2}\ell n|x+3|+C$

2. **(i)** $\frac{-1}{3}\ell nx+\frac{2}{33}\ell n\left|x-\frac{3}{2}\right|+\frac{3}{11}\ell n\left|x+\frac{1}{3}\right|+C$ **(ii)** $\frac{-1}{x+1}+\frac{1}{2(x+1)^2}dx$

3. **(i)** $\frac{-4}{x-2}-\frac{11}{2(x-2)^2}+C$ **(ii)** $2\ell n\left|\frac{x+4}{x+2}\right|-\frac{5x+12}{x^2+6x+8}+C$

4. **(i)** $\ell n|x+2|+\frac{3}{x+2}+C$ **(ii)** $\frac{-1}{20}\ell n|x^2+1|+\frac{3}{10}\tan^{-1}x+\frac{1}{10}\ell n|x+3|+C$

5. **(i)** $-\frac{1}{2}\frac{1}{2}\ell n\left|\frac{x-1}{x+1}\right|+\frac{1}{2}\tan^{-1}x+C$ **(ii)** $4\ell n|x+1|+\frac{1}{x+1}-4\ell n|x+2|+C$

6. **(i)** $=\frac{1}{3}\ell n|x-1|-\frac{1}{6}\ell n|x^2+x+1|-\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+C$

(ii) $\frac{1}{15}\left[\frac{1}{2}\ell n\frac{(t-1)^2}{t^2+t+1}-\sqrt{3}\tan^{-1}\frac{2t+1}{\sqrt{3}}\right]+C$

7. **(i)** $\frac{x^2}{2}\ell nx-\frac{1}{2}.\frac{x^2}{2}+C$ **(ii)** $-x.e^{-x}-e^{-x}+C$

8. **(i)** $\left(x^2-1\right)e^{x^2}+C$ **(ii)** $\frac{1}{2}\left[-x^2\cos x^2+\sin x^2\right]+C$

9. **(i)** $-\frac{1}{x}\left[(\ell nx)^2+2\ell nx+2\right]+C$ **(ii)** $-2\sin^{-1}\sqrt{x}.\left(\sqrt{1-x}\right)+2\sqrt{x}+C$

10. **(i)** $x.\tan^{-1}x+\ell n\sqrt{1+x^2}+C$ **(ii)** $x\tan^{-1}x-\left(\tan^{-1}x\right)^2-\ell n\sqrt{1+x^2}+\frac{\left(\tan^{-1}x\right)^2}{2}+C$

11. **(i)** $x\tan^{-1}\sqrt{x}-\sqrt{x}+\tan^{-1}\sqrt{x}+C$ **(ii)** $x\ell n\left(x+\sqrt{1+x^2}\right)-\sqrt{1+x^2}+C$

12. **(i)** $\frac{\sec x\tan x}{2}+\frac{1}{2}\ell n|\sec x+\tan x|+C$ **(ii)** $\frac{x}{\sqrt{1-x^2}}\sin^{-1}x+\ell n\sqrt{1-x^2}+C$

13. **(i)** $3\left(x^2-2\right)\sin x-\left(x^3-6x\right)\cos x+C$ **(ii)** $\frac{1}{27}\left[\left(9x^2-18x+25\right)\sin 3x+3(2x-2)\cos 3x\right]+C$

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RACE
Regular Analysis Through Continuous Exercise

Indefinite Integration : RACE-08

Subjective Type Question

Evaluate the following integrals :

1. **(i)** $\int \frac{e^x}{x}(1+x.\ln x)\,dx$

MII351

(ii) $\int e^x\left(\tan^{-1}x+\frac{1}{1+x^2}\right)dx$

MII352

2. **(i)** $\int \frac{e^x(x-1)}{(x+1)^3}dx$

MII353

(ii) $\int xe^{x^2}\left(\sin x^2+\cos x^2\right)dx$

MII354

3. **(i)** $\int \frac{e^{2x}(\sin 4x-2)}{1-\cos 4x}dx$

MII355

(ii) $\int e^x\left[\ln(\sec x+\tan x)+\sec x\right]dx$

MII356

4. **(i)** $\int \frac{e^{\sin^{-1}x}}{2x\sqrt{x-x^3}}\left(2x-\sqrt{1-x^2}\right)dx$

MII357

(ii) $\int(\ell nx+1)dx$

MII358

5. **(i)** $\int \sin(\ell nx)dx$

MII359

(ii) $\int\left[\ln(\ln x)+\frac{1}{(\ln x)^2}\right]dx$

MII360

6. **(i)** $\int \frac{\ln x}{(\ln x+1)^2}dx$

MII361

(ii) $\int e^{x/2}\sin\left(\frac{x}{2}+\frac{\pi}{4}\right)dx$

MII362

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7. **(i)** $\int e^{-2x}\sin 3x dx$

MII363

(ii) $\int e^{\sin x}\frac{x\cos^3 x-\sin x}{\cos^2 x}dx$

MII364

8. **(i)** $\int\frac{(x+\sin x)^2+(1+\cos x)^2}{(x+\sin x)(1+\cos x)}\,dx$

MII365

(ii) $\int\frac{2x-\sin 2x}{1-\cos 2x}dx$

MII366

(iii) $\int\left(3x^2\tan\left(\frac{1}{x}\right)-x\sec^2\left(\frac{1}{x}\right)\right)dx$

MII367

9. **(i)** If $I_n=\int\frac{x^n dx}{\sqrt{x^2+a}}$, then prove that $I_n+\frac{n-1}{n}a.I_{n-2}=\frac{1}{n}.x^{n-1}.\sqrt{x^2+a}$

MII368

(ii) If $I_n=\int x.\text{cosec}^n x.dx$, n $\geq$ 3, then prove that

$I_n=-\frac{1}{n-1}.\cos ec^{n-2}x.\left(x\cot x+\frac{1}{n-2}\right)+\frac{n-2}{n-1}.I_{n-2}+\frac{C}{n-1}$

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MII369

(iii) Prove that $\int\sin^p x\cos^q x dx=\frac{\sin^{p+1}x.\cos^{q+1}x}{p+q}+\frac{q-1}{p+q}\int\sin^p x\cos^{q-2}x dx$

MII370

10. **(i)** $\int\frac{dx}{\sqrt[4]{(x-1)^3(x+2)^5}}$

MII371

(ii) $\int\frac{dx}{(x+1)^{6/5}(x-3)^{4/5}}$

MII372

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ANSWER KEY

1. **(i)** $e^x.\ell nx + C$ **(ii)** $e^x \tan^{-1} x + C$

2. **(i)** $\frac{e^x}{(x+1)^2}+C$ **(ii)** $\frac{1}{2}e^{x^2}\sin(x^2)+C$

3. **(i)** $\frac{1}{2}e^{2x}\cot 2x+C$ **(ii)** $e^x \ell n(\sec x + \tan x) + C$

4. **(i)** $\frac{e^{\sin^{-1}x}}{\sqrt{x}}+C$ **(ii)** $x\ell nx + C$

5. **(i)** $\frac{x}{2}(\sin \ell nx-\cos \ell nx)+C$ **(ii)** $x\left(\ell n(\ell nx)-\frac{1}{\ell nx}\right)+C$

6. **(i)** $\frac{x}{\ell nx+1}+C$ **(ii)** $\frac{e^{x/2}\sin\frac{x}{2}}{\sqrt{2}}+C$

7. **(i)** $I=-\frac{e^{-2x}}{13}(3\cos 3x+2\sin 3x)+C$ **(ii)** $e^{\sin x}(x-\sec x)+C$

8. **(i)** $x\tan\frac{x}{2}+\ell n|x+\sin x|+C$ **(ii)** $-x\cot x+C$ **(iii)** $x^3\tan\frac{1}{x}+C$

9. **(i)** $I_n=\frac{x^{n-1}\sqrt{x^2+a}}{n}-\frac{(n-1)}{n}aI_{n-2}$ **(ii)** $I_n=\left(\frac{n-2}{n-1}\right)I_{n-2}-\frac{\text{cosec}^{n-2}x}{n-1}\left(x\cot x+\frac{1}{n-2}\right)+\frac{c(n-2)}{n-1}$

(iii) $I=\frac{(\cos x)^{q-1}(\sin x)^{p+1}}{p+q}+\frac{q-1}{p+q}\int(\cos x)^{q-2}(\sin x)^p\,dx$

10. **(i)** $\frac{4}{3}\sqrt[4]{\frac{x-1}{x+2}}+C$ **(ii)** $\frac{5}{4}\left(\frac{x-3}{x+1}\right)^{1/5}+C$

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